

Development of an Evaporation-Based Preconcentrator for Trace Detection in Defense Applications

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Background

Trace detection allows for the early identification of a wide range of threats, including biological agents like anthrax, toxic industrial chemicals such as benzene and Per- and polyfluoroalkyl substances (PFAS) that can contaminate water and soil.



Preconcentration is essential to enhance the detectability and accuracy of analytical measurements, especially when dealing with samples that contain very low concentrations of the target analyte.

Evaporation is a simple and efficient preconcentration method that removes solvent, preserves the integrity of the analyte, and is easy to control and monitor.

Designing the device to be compact allows it to be easily integrated with portable detection instruments, enabling quick, on-site analysis, which is crucial for defense applications.

Experimental Setup

The proposed structure of an evaporator is a 'sandwich' configuration where a porous separator membrane is positioned between a liquid channel and a gas channel.

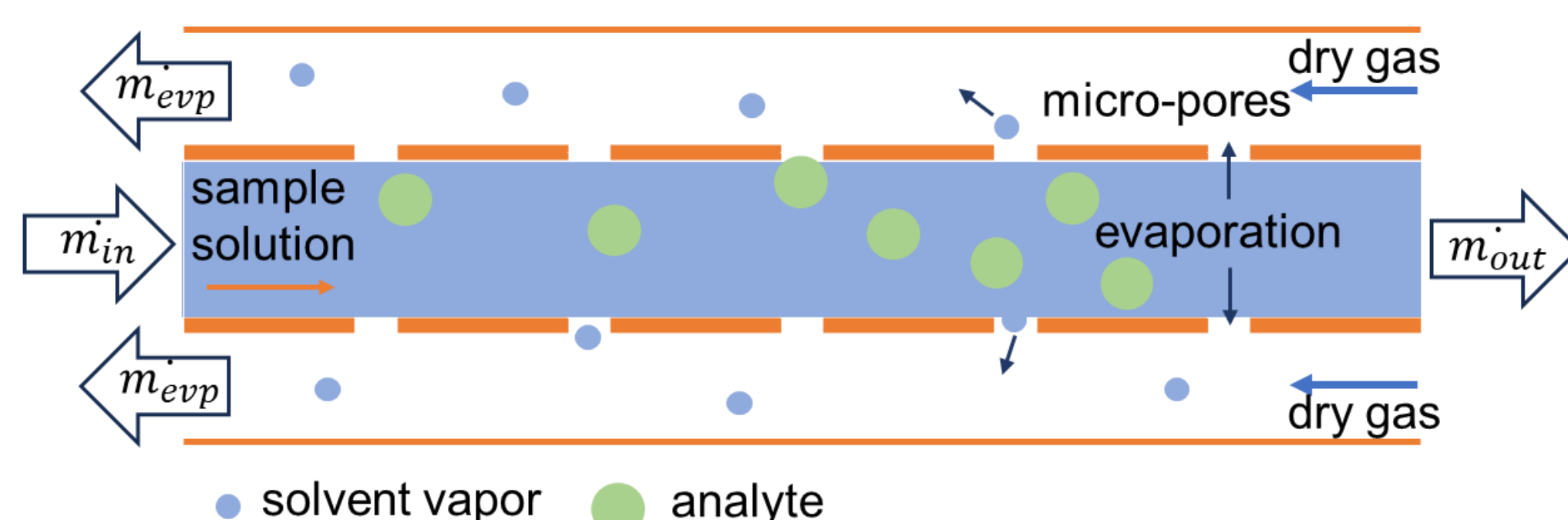


Figure 1. 2D schematic of evaporation-based preconcentrator.

An evaporation-based preconcentrator should have the following properties:

- Solvent vapor permeability: The membrane should allow solvent vapor to pass through easily.
- Analyte retention: Analyte particles must be unable to pass through the membrane.
- Pressure tolerance: The pores of the membrane should be capable of withstanding sufficient pressure for effective operation.

A micro-3D printed device is shown below:

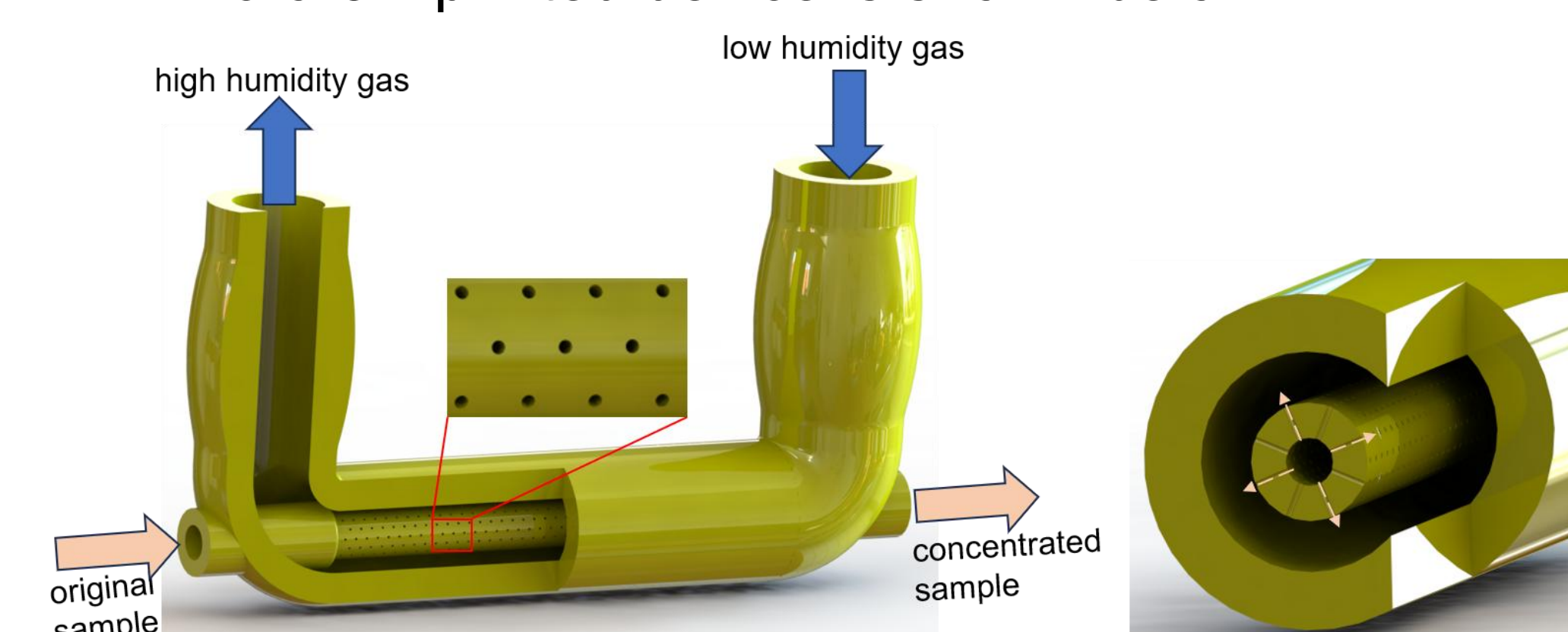


Figure 2. Image of a 3D printed micro-pore preconcentrator.

- In this design, the micropores have a diameter of 50 μm , with an adjacent spacing of 250 μm between each pores.

The evaporation system:

- A syringe pump is connected to the evaporator as the inlet, while the outlet leads to a sealed collection vial placed in ice to minimize evaporation during the collection process.
- A heater plate is attached to the underside of the heating box, accompanied by a temperature on-off controller. This setup allows the temperature to be maintained within $\pm 0.2^\circ\text{C}$ of the target value.
- The humidity sensor is positioned downstream of the evaporation process. It continuously monitors the relative humidity of the outlet gas, providing an assessment of the evaporation process.

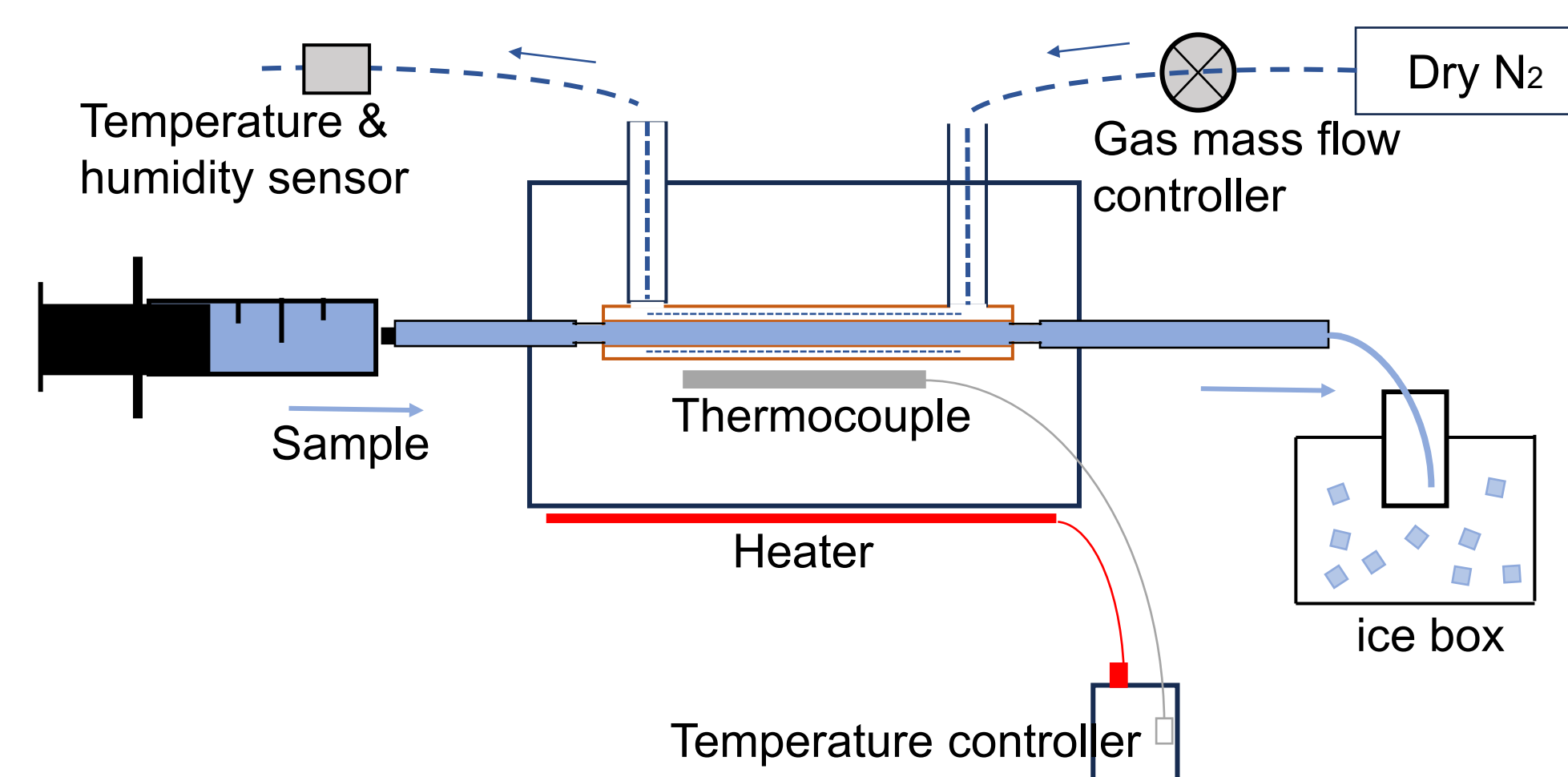


Figure 3. Overall schematic of the evaporation preconcentrator system.

Theoretical calculations:

evaporation ratio (ER) and concentration factor (CF)

$$ER = C \phi \frac{M_{WV}}{M_{SG}} \frac{P_{WV,SAT}}{P_{SG}} \frac{\dot{m}_{SG}}{\dot{m}_{LS}} \quad CF = \frac{1}{1 - ER}$$

C : a constant empirical coefficient

ϕ : the relative humidity increase from inlet to outlet of the sweeping gas

M_{WV} : the molar masses of water vapor ; M_{SG} : the molar masses of sweeping gas

$P_{WV,SAT}$: the saturation pressures of water vapor

P_{SG} : the pressure of water vapor in sweeping gas

$\dot{m}_{SG}$: the mass flow rates of sweeping gas; $\dot{m}_{LS}$: the mass flow rate of liquid solution

Results and Discussion

Effect of evaporator parameters:

- Temperature: higher temperatures result in faster evaporation, but if too high, could damage the compound of interest.
- Liquid flow rate: higher liquid flow rate leads to a lower evaporation
- Gas flow rate: higher gas flow rates result in faster evaporation.

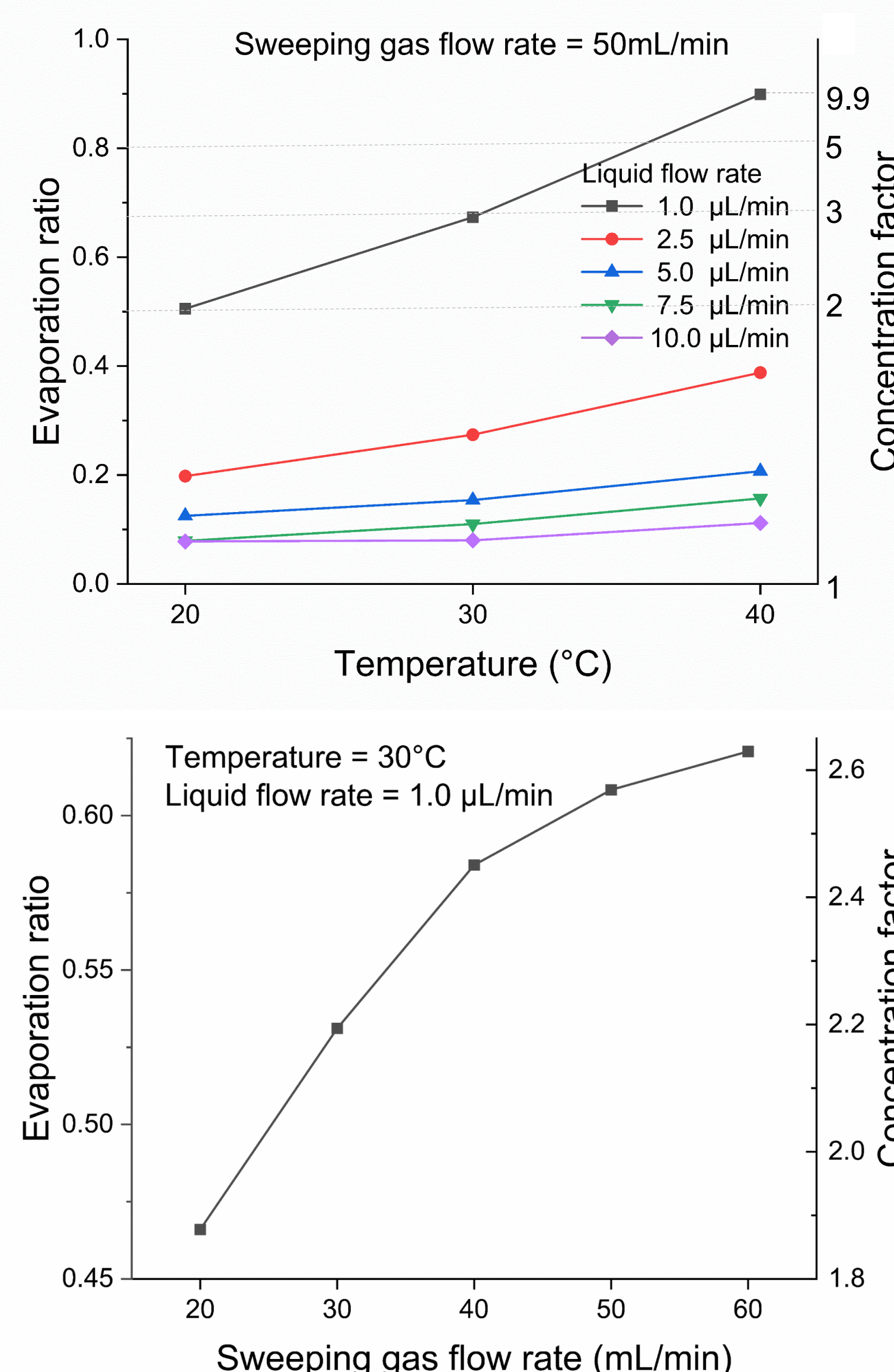


Figure 4. Evaporation ratios and concentration factors for different experimental parameters.

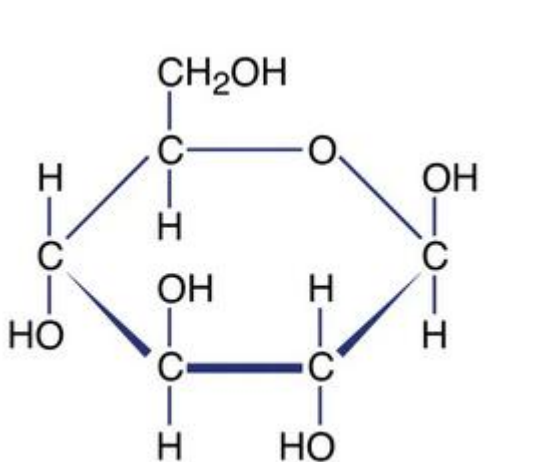
Sample concentration quantification:

- FTIR (Fourier Transform Infrared Spectroscopy) measures the absorption of infrared light by a sample, identifying molecular vibrations at specific frequencies.
- Each chemical compound has unique absorption peaks, allowing for identification and quantification.
- The intensity of these peaks correlates with the concentration, enabling quantification.

Examples:

Glucose

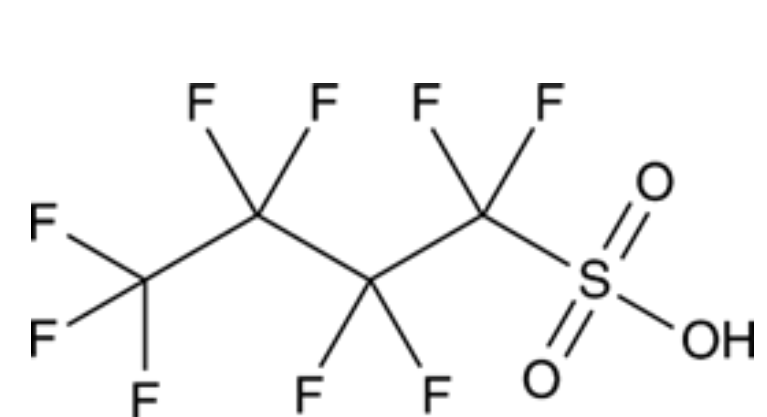
Glucose was dissolved in distilled water, and different concentrations ranging from 10 to 100 mmol/L were prepared through serial dilution.



	10 mmol/L glucose	10 mmol/L glucose
Liquid flow rate (uL/min)	2.5	1
Gas flow rate (mL/min)	50	50
Temperature (°C)	30	40
ER	0.23	0.89
CF	1.3	8.8

PFAS

Nonafluoro-1-butanesulfonic acid, also known as PFBS was dissolved in distilled water, and different concentrations ranging from 100 to 5000 ppm were prepared through serial dilution.



	250 ppm PFBS	100 ppm PFBS
Liquid flow rate (uL/min)	3	1.2
Gas flow rate (mL/min)	50	60
Temperature (°C)	42	42
ER	0.2	0.98
CF	1.25	48

- Liquid flow rate, gas flow rate and temperature have direct effect on the evaporation rate.
- These parameters can be optimized for a particular application.

Conclusions

- A novel evaporation-based preconcentrator has been shown to provide a concentration increase of 8.8 times with glucose and 48 times with PFAS.
- Experimental results validate the significant impact of temperature, liquid flow rate, and gas flow rate on evaporation efficiency and concentration factor, providing valuable insights for optimization.
- This system demonstrates strong potential for portable, rapid trace detection of low-concentration analytes in defense and environmental monitoring applications.

Acknowledgment

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* Authors are co-inventors on a patent application relating to this technology, owned by The University of Texas at Austin.